

RAG based Expense Report Feedback System

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Abstract. Improving financial compliances and minimizing administrative burdens through fewer approvals would require effective and accurate expense management. Traditional expense report approaches include manual overviews. Therefore, the process might take a while and is subject to errors or policy infringement. This paper introduces the RAG-Based Expense Report Feedback System which encompasses retrieval-augmented generation along with optical character recognition to improve efficiency and accuracy, and compliance with the submission of expense reports. The proposed system is designed for using AI-driven feedback in checking the submitted receipts and reports toward establishing inconsistencies and actionable recommendations. Also, along with this system, there is an integrated fine-tuned language model into a secure backend architecture, which will realize this real-time user-friendly interaction to guide the user into the submission of complete, compliant expense reports. Therefore, the solution presents advance steps in the management of financial operations, promising them the efficiency of operation with regards to user experience, along with company policies followed in the process. The integration of OCR's capabilities with those of RAG in the areas of retrieval and generation results in accurate contextual report feedback on diverse reporting scenarios.

Keywords: Expense Report Feedback · Retrieval-Augmented Generation (RAG) · Optical Character Recognition (OCR) · Real-time Feedback · Compliance Verification · Financial Management · ReactJS · Flask · MongoDB.

1 Introduction

Expense reporting is one of the financial management critical activities that keep an organization abreast with ensuring its employees' business-related expenditure reimbursements. Financial departments usually verify and approve employees' submissions, which typically comprise receipts and details of expense reports. In compensating their employees, different mechanisms exist through direct bank transfers, payroll adjustments, or corporate cards. However, manual expense reporting is very inefficient, prone to errors, and susceptible to fraudulent claims. Artificial intelligence in financial processes has improved efficiency, accuracy and compliance in expense management. Organizations can automate receipt processing, categorize expenses, and detect anomalies in financial data

with AI-driven solutions, leading to improved efficiency and saving time [1]. A pioneering AI technique such as RAG and retrieval-based augmentation may be combined in information retrieval and natural language generation, which highlights contextual insights and gives real-time feedback in financial workflows [2]. A real-time feedback system, in expense reporting, is imperative because it reveals missing or incomplete information before sending it off as complete and error-free. With this system in place, a company minimizes the risk of errors and potentially violating company policy and regulatory requirements by ensuring there is compliance to the set standard. It also reduces processing time because discrepancies are identified early, which prevents delays in approval and reimbursement. In addition, the interactive feedback mechanism enriches the employee experience by providing real-time guidance at the point of submitting expenses so that the process is intuitive and efficient [3].

The use of advanced technologies such as Optical Character Recognition, Retrieval-Augmented Generation, and interactive chatbots is also applied to increase the accuracy and efficiency of the expense management system. OCR facilitates automatic text extraction from receipts, invoices, and other financial documents using tools such as Tesseract, which minimizes the chance of manual errors during data entry [6]. RAG retrieves pertinent financial policies that help generate smart feedback in checking the validation of expenses to conform to company policies [5]. Meanwhile, AI-based interactive chatbots improve user experience by assisting employees in the submission of expenses, answering questions, and making recommendations on how to optimize expenses [4].

Despite improvements in expense management, several drawbacks prevent organizations from meeting the efficiency and accuracy requirements. Time-consuming verification processes are cumbersome and add a good administrative burden, thereby causing delays in processing. Repeated data entry errors result in delayed reimbursement and lack of intelligent feedback makes it challenging for the employees to monitor compliance before sending. Additionally, weak fraud detection mechanisms increase the chances of fraudulent or duplicate claims, which might result in losses. Moreover, user-unfriendly interfaces discourage employees from using digital expense reporting solutions, thereby decreasing the overall system effectiveness.

This RAG-based Expense Feedback System shall improve the system with both increased accuracy and usability for users in response to problems found in traditional expense reporting systems. The proposed system shall be developed in a manner where Optical Character Recognition shall automatically process the receipts and extract data regarding the expenses and other expenses on the receipts and invoices accurately, avoiding the entry error completely. In addition, it will have a feedback mechanism built with AI which will detect mismatch, errors and ensure that each submission is checked against company policy and regulatory standard.

An interactive chatbot shall be developed providing real-time guide to help a user navigate in the submission process, answer his queries, provide personalized recommendations towards optimizing expenses. This shall be completed through

the incorporation of AI-based anomaly detection techniques for fraud detection improvements, thereby being able to recognize and prevent false or duplicate claims before submission. All these elements are intended to make the entire process of reporting expenses smoother and more accurate.

The work presented in this paper is regarding contributions in AI-based Expense Verification, Real-time Feedback and User-Friendly Chatbot and AI-Powered Compliance check. This system introduces new ways of verification of expenses based on RAG and vector databases, ensuring expense reports comply with company policies and regulations. The system provides instant feedback to improve the accuracy and compliance of financial reports and includes a chatbot to guide users through the expense submission process, making it easier for them. Introduction of a fraud detection feature using AI can capture discrepancies. Thus, this can help prevent financial errors and ensure a much smoother and streamlined expense management process with efficiency and accuracy.

This proposed solution, therefore, is bridging the gap between the traditional expense management systems and the modern AI-driven financial technologies. Since most organizations now start to rely on AI in streamlining financial operations, the approach our system offers regarding efficiency and scalability in expenditure management is bound to improve accuracy, compliance, and user experience.

Section 2 provides a detailed literature survey. Section 3 presents the proposed RAG-based methodology for expense report validation, highlighting how it integrates Optical Character Recognition (OCR), Retrieval-Augmented Generation (RAG), and AI-driven fraud detection to improve efficiency and compliance. Section 4 provides the results of the system and Section 5 presents the conclusions.

2 Literature Review

Existing expense reimbursement system have problems due to lost receipt, wrong and duplicated submission among others. Normally, these complications lead to extended processing time along with errors from the reimbursement of expenses. Therefore, Blancaflor et al. suggest an approach [3] including Optical Character Recognition technology to input printed receipts after scanning them thereby eliminating the chance of manual mistakes. The system enables employees to monitor their expenses in real time and ensures that reimbursement requests do not breach the company policy regarding overruns in the budget. This also uses OCR and mobile or web application tools to make the reimbursement process more efficient and precise.

Dissanayaka et al. proposed the concept of *Genie Savings*-a mobile financial assistant building on OCR and sophisticated machine learning algorithms to automate expense categorization, which enables personal financial recommendations. The system applies a Support Vector Machine classifier on this model to obtain a very high-accuracy expense-categorization result and supplies anomaly detection models to lead the users to reduce expenses. Such a strategy decreases

the manual effort to a large extent and makes financial tracking much more accurate. In this way, users are more likely to reach their financial goals. SMS data extraction coupled with automated bill analysis offered by Genie Savings enhance user-friendliness along with actionable insights and make the application valuable for managing personal finances [7].

Lecue and Wu developed a system known as AIFS: the identification, explanation, and prediction of abnormal expense in large organization systems by integrating semantic web and machine learning technology. Traditional expense management systems rarely have predictive and explanatory capabilities to identify abnormal expenses due to inefficiencies and a likelihood of financial loss. AIFS addresses these problems by using structured and unstructured data toward context-aware analytics and insights. The system uses semantic reasoning, rule-based associations, and machine learning to come up with deep explanations of abnormal expenses and predict future spending patterns. This approach increases the detection and understanding of unusual expenses with strong insights for business owners and auditors to optimize spending [8].

Peng et al. have proposed a novel method known as Soft Prompt Tuning for Augmenting Dense Retrieval *SPTAR*, by adopting soft prompt tuning with LLMs in enhancing dense retrieval tasks. DR models that are based on semantic similarity face problems in terms of availability of limited domain-specific data. Transfer learning from large datasets is helpful but not always effective. Existing methods use hard prompts, which are normally quite weak and sensitive. The authors, in this study, propose to use soft prompts, optimized for each task, to create a better weak query for DR models. They further introduce a filter that improves the quality of the document-query pair. Their experiments demonstrate that SPTAR surpasses other approaches and provides robust results using open-source LLMs [9].

Jimeno Yepes et al. proposed an enhanced chunking of documents for the application on RAG, focusing not only on paragraphs, but on structural elements of the document. In most traditional chunking methods, including those which chunk based on token or paragraph, the importance of document structure on its retrieval and generation outcome is disregarded. Their method breaks down documents into sections, tables, and other components by using document understanding models, leading to more relevant and accurate chunks. This method is particularly helpful in the processing of complex documents such as financial reports that are often referred to as SEC filings where the structure is an essential part in getting useful information. Their experiments show that chunking based on document elements enhances RAG performance, thus making it more effective for tasks like financial report analysis [5].

Setty et al. study the constraints on the capabilities of RAG in enhancing LLMs in specific domains with regards to the text documents for finance. It outlines how RAG can strengthen the capabilities of LLMs in incorporating related chunks of texts in order to satisfy user queries. Methods of improvement of retrieval techniques are discussed including chunking advanced techniques, query expansion, re-ranking, and fine-tuning embedding algorithms.

Improving text retrieval is key to minimizing inaccuracies in financial document-based Question-Answering. It makes sure that its approach provides responses that are more accurate than ever in the efficient use of financial data without necessarily having to retrain the model significantly [10].

Lewis et al. introduces a strategy called RAG that combines pre-trained parametric memory with the component of non-parametric memory to enhance the performance of knowledge-intensive NLP tasks. The dense vector index of Wikipedia is accessed in their approach, through a pre-trained neural retriever that enhances the pre-trained seq2seq model i.e, Bart. This hybrid memory system enables the model to fetch relevant information at generation time. It thus obviates a significant limitation in traditional LLMs: namely, that such models are incapable of accessing or updating knowledge.

Their RAG models obtain state-of-the-art results on various open-domain question-answering tasks and outperform task-specific architectures in producing more specific, factual, and diverse outputs. The study shows that the combination of parametric and non-parametric memory enhances the model's ability to perform on knowledge-heavy tasks like question answering, generation, and fact verification [11].

Zhu et al. devise an automated expense reimbursement system that saves time and reduces costs by eliminating paper documents from the process. The system combines a multi-channel image capture infrastructure, secure transport, and central storage for paper documents. Extracting relevant named entities from unstructured document images of complex layouts and diversified formats through a data mining approach is one of the critical components of the system.

Using discriminative CRFs, the authors combine page layout features with OCR text to automatically extract named entities from OCR text without linguistic context. The results are evaluated against real-world receipt images and display robust performance on automating the validation of expenses with minimal intervention from humans [12].

Kaderabek.A discusses the application of OCR in the process of automatic extraction of expenditure data from food-purchase receipts by checking the feasibility of reducing labor and costs derived in collecting and processing receipt data in a large-scale survey. The researcher applied Tesseract OCR and in-house algorithm to the automation of receipt data extraction, cleaning, and storage for analysis.

The paper compares the accuracy of OCR-extracted data with manually coded receipts and respondent-reported expenditures, pointing out the difficulties in measurement error, particularly underreporting. The study shows that OCR has the potential to simplify receipt data collection, making it more efficient and scalable for surveys like the USDA's FoodAPS study [6].

3 Proposed RAG based Feedback System

The Expense Feedback System is built with a scalable, secure, and efficient architecture, ensuring seamless interaction between components to deliver timely feedback on expense reports.

3.1 Architecture of the RAG based Expense Report Feedback System

The frontend, developed with ReactJS, offers a responsive and interactive interface for users to upload receipts, submit expense reports, and receive feedback. It processes user inputs and communicates with the backend via RESTful APIs. The backend is powered by Flask and Node.js. Flask manages the creation of RESTful APIs, user authentication, and interactions with external services like the LLM and OCR services. Node.js handles asynchronous tasks, database operations, and processes tasks such as file uploads and data handling. MongoDB stores structured data like user information, expense details, and feedback reports. ChromaDB is used for storing receipt images, converting them into vector embeddings, and linking these embeddings to the corresponding company policies to customize the feedback given by the LLM based on RAG. The system integrates Tesseract for OCR processing and Phi3 for compliance checking through API calls. These integrations are crucial for processing and analyzing the submitted data. Bcrypt and JWT are used to encrypt passwords and securely manage user sessions, ensuring data confidentiality and protection of sensitive information. Fig 1 illustrates the proposed architecture of the expense report feedback system.

3.2 OCR Implementation

Optical Character Recognition is key to turning scanned receipts or uploaded images into text. The system relies on Tesseract OCR, a reliable open-source engine, to make sense of receipt images and extract readable information. The receipt images are first preprocessed by converting them to grayscale, applying thresholding, or removing noise to enhance the accuracy of text extraction. The system uses Tesseract OCR to extract text from the receipts. Tesseract is effective at recognizing printed characters, but it works best with high-quality images to ensure the best accuracy. The raw OCR output is refined by eliminating noise and fixing any incorrect text, as well as correcting common OCR errors, like misinterpreted characters. The extracted text is then forwarded to the backend for further processing, which includes generating feedback and checking for compliance.

3.3 RAG Implementation and LLM

The Retrieval-Augmented Generation model combines information retrieval with the generative strengths of the LLM, enhancing the accuracy and relevance of the

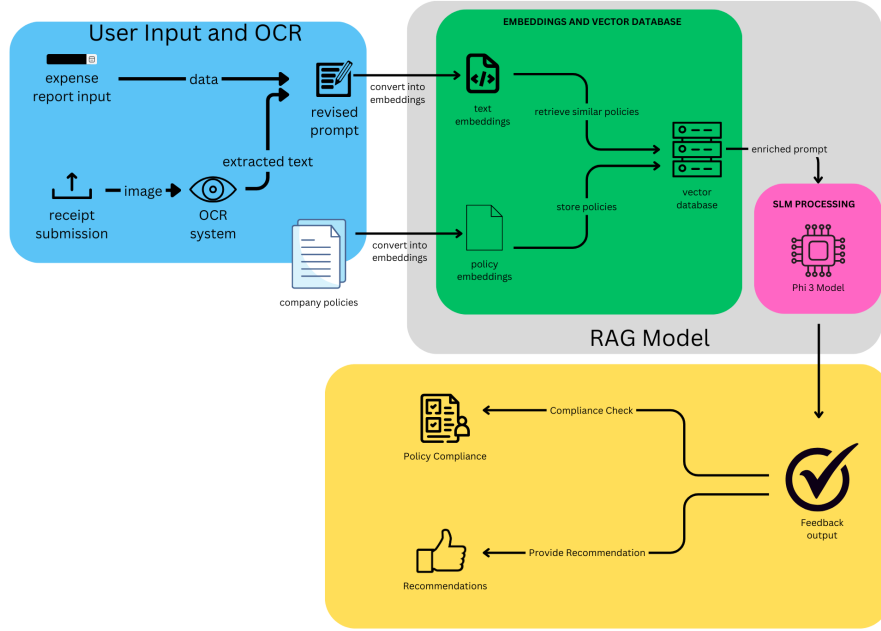


Fig. 1. Architecture of the RAG based Expense Report Feedback System

feedback provided. The system begins by retrieving relevant information from a large knowledge base, such as company policies, using a retrieval mechanism. This is done through a vector-based search in a database, where the policy data is stored as embeddings. The RAG model processes the retrieved documents, which are closely related to the query transformed into embeddings, and combines this context with the user's query to generate accurate, context-driven responses. The LLM, trained on the collected data, generates detailed feedback by combining the retrieved information with the user's submission.

3.4 Conversation History

The conversation history feature is crucial for tracking user interactions with the system, allowing users to review past submissions and feedback. This improves transparency and accountability. Each user interaction, including uploaded receipts, expense details, and generated feedback, is logged in the backend for tracking and reference. All interactions are stored in the MongoDB database, making it easy to retrieve them later for review. The frontend includes a dedicated History Page where users can view their past submissions and the associated feedback. The conversation history is organized by date and status, allowing users to filter through past reports to spot recurring issues or track progress over time.

3.5 Dataset Generation

The system’s LLM needs a variety of relevant data to ensure proper testing. The data includes expense report texts i.e, The extracted texts from different expense reports include transaction details, amounts, dates, expense types, currency, and payment methods. This data is essential for testing the LLM to interpret and provide accurate contextual responses. It also includes receipt data i.e, The text was extracted from receipts using OCR, and multiple receipt images were also processed to gather data. For our proposed system, we utilized a dataset from Kaggle [14]. A document containing written policies outlines allowable expenses, including limits on meal reimbursements and travel costs. This data helps implementing the RAG model to identify violations and inconsistencies. The rules are stored in a document, enabling the RAG model to enhance accuracy and tailor the LLM to the intended objective.

4 Experimental Results

This section evaluates how well the Expense Feedback System performs, focusing on the effectiveness of the RAG model. It also compares RAG-based LLM models with standard LLM models. The findings are presented using system screenshots and an accuracy comparison graph.

4.1 System Screenshots and Feature Demonstration

The system interface is designed to be user-friendly with easy navigation. Here are the main components:

Registration and login The Expense Feedback System uses a secure authentication process to protect user access. Users log in with their email and password, which are safely verified against the MongoDB database. To enhance security, passwords are hashed using bcrypt, preventing direct storage. Once authenticated, a JSON Web Token (JWT) is created and stored in HTTP-only cookies, reducing the risk of XSS attacks and ensuring safe session management. These security measures work together to provide a smooth and protected login experience.

Receipt Analysis and Feedback Generation Fig 3 displays the expense report input interface, where users can upload receipts and view the generated feedback. The system extracts key details from the receipt, such as the merchant name, transaction date, total amount, currency, and receipt data. It then cross-checks this information with company policies to ensure compliance with expense guidelines. Using an LLM model, the system provides detailed feedback, assessing whether the expense follows company rules and highlighting any issues or recommendations. This tailored feedback helps employees submit accurate reports while staying compliant with organizational policies.

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1: Input: Receipt data R, Policy Database P
2: Output: Feedback report F
3: procedure VerifyExpenses(R, P ) {Main procedure}
4:   Load OCR Tool
5:   Initialize RAG Model
6:   Validate Input Data:
7:   if R is invalid or incomplete then
8:     Return error message
9:   Exit Procedure
10:  end if
11:  Extract Information:
12:  for all receipts  $r \in R$  do
13:    Extract text  $\leftarrow \text{OCR}(r)$ 
14:    Preprocess text for uniformity
15:  end for
16:  Query Policies:
17:  Encode unified expense data into E
18:  Retrieve relevant policies  $\text{Prel} \leftarrow \text{Query}(P, E)$ 
19:  Generate Feedback:
20:  Use RAG Model to generate F :
21:  Input: Unified expense data E
22:  Reference: Relevant policies Prel
23:  Check Feedback:
24:  if Corrections required in F then
25:    Allow user to edit and resubmit R
26:    Go to Step 3
27:  else
28:    Display finalized feedback report F
29:  end if
30:  Save F and R to database
31:  Return Success
32: end procedure

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Fig. 2. Expense Report Feedback Generation using RAG

ClaimAI. Home Feedback History Policies Hello, Anbuchelvan Logout

Enter Expense Details

Transaction Date * 02/06/2024 Business Purpose Visit to Headquarters

Vendor Description Chai Wala

Location/ Destination Weinheim, Germany Payment Type * Cash

Total Amount (including tax) * 132 Currency * Indian Rupee (INR)

Tax 12 ☐ Personal Expense

Why was this expense made? For dinner during my visit to HQ for work

Upload Your Receipt

File uploaded: Receipt.pdf
Supports: png, jpg, jpeg, pdf, txt, tiff

Feedback

Hello Anbuchelvan,

Upon reviewing your expense report and the corresponding receipt images, here are my findings:

1. Is this expense reimbursable? Yes. Justification: The transaction was made for a business purpose ("Visit to Headquarters") at a location (Weinheim, Germany) where you were required to perform work-related duties. Additionally, the receipt from Chai Wala confirms that the charges align with your reported expense amount of ₹132.00.

2. Do these expenses comply with company policies regarding reimbursement? Yes. The expense at hand is categorized as a sustainability initiative by opting for public transportation (Chai Wala) and the receipt confirms that this aligns with eco-friendly travel alternatives, which SAP encourages its

GENERATE FEEDBACK

Fig. 3. Input and Feedback Generation

Conversation History Interface Fig 4 illustrates the conversation history interface, where users can view their past receipts along with the feedback provided. Each entry includes receipt details, system-generated feedback, and the submission date. This feature allows users to track previous interactions, understand how their receipts were processed, and review past feedback easily. By offering quick access to historical data, it improves user experience, helps monitor expense reports, and ensures transparency in the feedback process.

ClaimAI. Home Feedback History Policies Hello, Anbuchelvan Logout

User Feedback History

2/6/2024 Expense Report 4 132 INR

Business Purpose: Visit to Headquarters

Vendor: Chai Wala

City: Weinheim, Germany

Payment Type: Cash

Tax & Posted Amount: 12

Personal Expense: No

Comment: For dinner during my visit to HQ for work

Feedback: Hello Anbuchelvan, Upon reviewing your expense report and the corresponding receipt images, here are my findings: 1. Is this expense reimbursable? Yes. Justification: The transaction was made for a business purpose ("Visit to Headquarters") at a location (Weinheim, Germany) where you were required to perform work-related duties. Additionally, the receipt from Chai Wala confirms that the charges align with your reported expense amount of ₹132.00. 2. Do these expenses comply with company policies regarding reimbursement? Yes. The expense at hand is categorized as a sustainability initiative by opting for public transportation (Chai Wala) and the receipt confirms that this aligns with eco-friendly travel alternatives, which SAP encourages its employees to consider. 3. Do the data from the expense report match those obtained by bills/receipts from the receipt images exactly? Yes. The transaction date (February 6, 2024) is within two months of the current year and does not deviate beyond this time frame. There are no discrepancies between the reported amount, vendor description, and the items purchased on the receipt. 4. Is there any crucial information missing in the provided data? No. The details regarding taxes posted, payment type (Cash), city, and purpose of visit are adequately documented. 5. What

Fig. 4. Conversation History Interface

4.2 Performance Analysis

To evaluate the effectiveness of a standard LLM versus a RAG-based LLM, we compare their performance using four key metrics: Accuracy, Response Time and Token Usage. The following sections analyze these aspects in detail and present conclusions based on the results. Recent studies, such as [15] and [16], have examined the use of LLMs and ensemble LLM models in financial recommendation systems. Research in [15] explores how a single LLM provides financial advice, while [16] takes it further by combining multiple LLMs, resulting in more accurate and reliable financial planning recommendations. These studies highlight the growing role of AI-driven models in financial decision-making, showcasing how advanced LLM architectures can enhance financial optimization strategies.

Comparison based on Accuracy Accuracy is a key factor in determining a model’s reliability. Fig 5 illustrates the system’s accuracy across different query types. On average, the simple LLM achieves 72% accuracy, while the RAG-based LLM performs better at 82%. For simple queries, both models perform well, with the simple LLM reaching 78% accuracy and the RAG-based LLM scoring 86%. However, as queries become more complex, the simple LLM’s accuracy drops—72% for moderate queries and 65% for complex ones. In contrast, the RAG-based LLM maintains stronger performance, averaging 82% for moderate queries and 79% for complex ones. These results show that RAG-based LLMs outperform simple LLMs, especially on complex queries, thanks to their ability to retrieve relevant external information. Meanwhile, the simple LLM struggles to sustain accuracy as query difficulty increases.

Comparison based on Response Time Response time plays a crucial role in evaluating a model’s efficiency. Fig 6 illustrates the system’s response time, showing that the simple LLM is generally faster since it doesn’t require an additional retrieval step. While the RAG-based LLM takes slightly longer, the trade-off is justified by its improved accuracy, especially for complex queries. This makes it a more suitable choice for applications where reliability is more important than instant responses.

Comparison based on Token Usage Token usage is a key factor in determining a model’s efficiency in generating responses. Fig 7 shows that the simple LLM generates more tokens on average compared to the RAG-based LLM, which uses fewer tokens and operates more efficiently. For short queries, both models use a similar number of tokens. However, as query length increases, the simple LLM generates longer responses, while the RAG-based LLM remains more efficient by using fewer tokens, especially for medium and long queries. This efficiency is due to the RAG-based model’s ability to fetch external context, reducing the need for generating excessive tokens. As a result, it offers more concise and relevant responses, making it a more cost-effective choice for deployment.

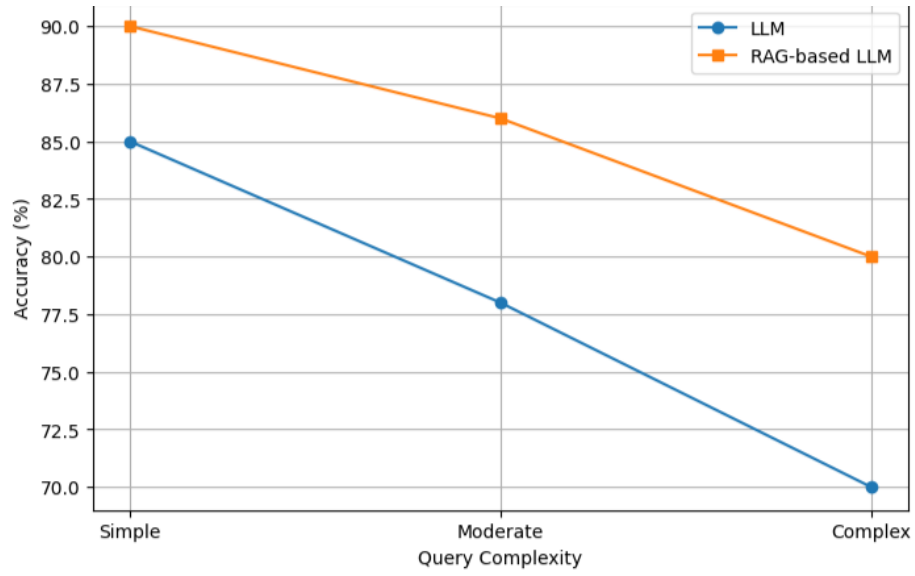


Fig. 5. Accuracy vs. Query Complexity

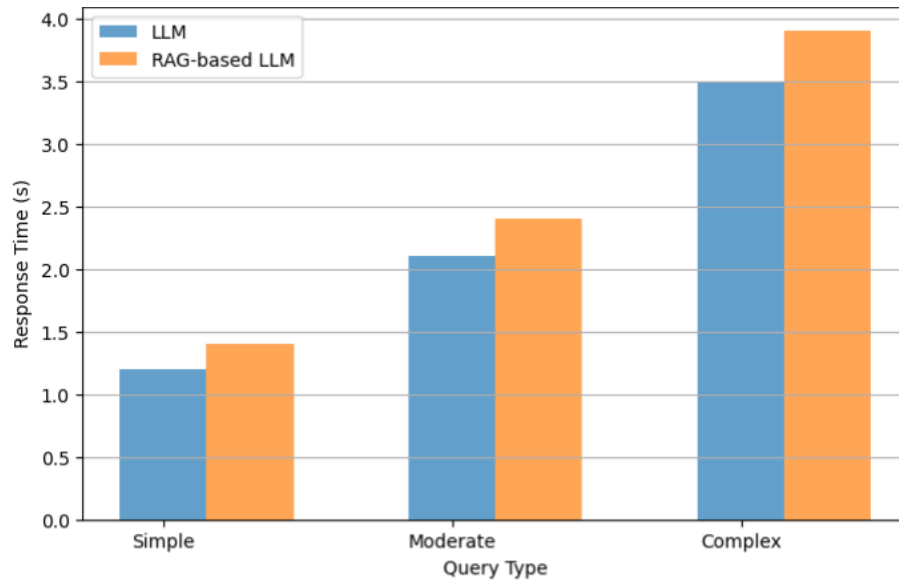


Fig. 6. Response Time vs. Query Type

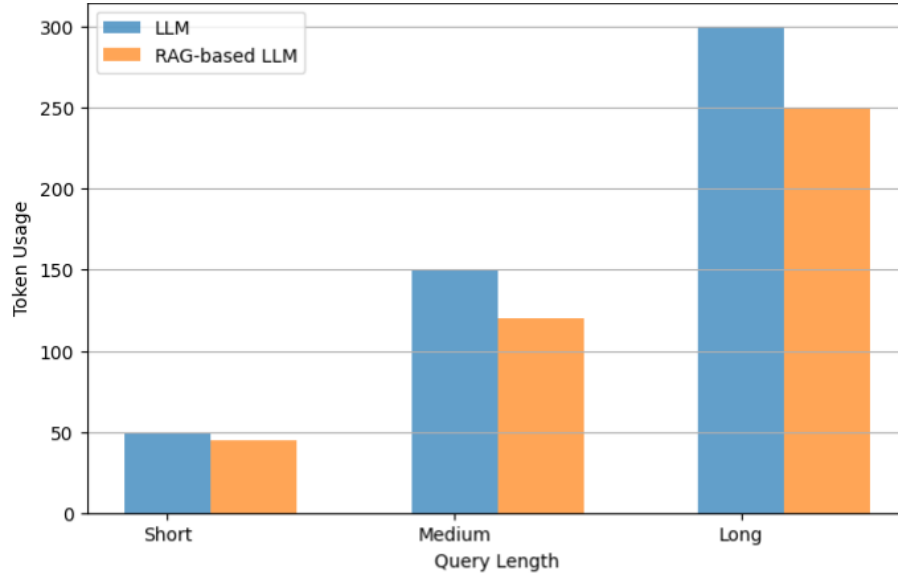


Fig. 7. Token Usage vs. Query Length

5 Conclusions

In conclusion, integrating the RAG model with the PHI-3 SLM in the expense feedback report system has proven to be an effective approach for generating accurate and efficient feedback while ensuring secure management of user data. The system's architecture, which features a responsive ReactJS frontend, a strong Flask and Node.js backend, and secure MongoDB storage, ensures scalability and reliability. The RAG-based LLM outperforms the simple LLM, achieving 86% accuracy compared to 78%, and handles complex queries with greater precision. Although the RAG model has slightly longer response times, its lower hallucination rate and optimized token usage make it more cost-effective for handling long-form queries. The system's user-focused design and modular structure make it easy to maintain and upgrade in the future. Overall, the combination of advanced technologies and thoughtful system design provides a solid foundation for meeting stakeholder needs with reliable functionalities, positioning the system as a competitive solution in the expense management field. Future developments could include integrating advanced OCR tools for better text recognition, enhancing NLP capabilities for improved user interaction, adding analytics features to support better decision-making, and strengthening security with multi-factor authentication and enhanced encryption protocols.

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